
Beyond Accuracy: Formalization for Understanding Information Exchange in Multi-Agent Debate

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Multi-agent debate allows language models to exchange reasoning and revise their
2 answers, but final accuracy does not reveal which properties of that exchange distinguish
3 successful debates. We introduce a claim-level observability framework
4 that translates explanations into attributed, temporally ordered propositions and
5 explicit reasoning relations. Using the controlled Knight–Knave–Spy domain, we
6 computationally evaluate claim truth and local logical support, quantify information
7 content, and track propositions across agents. Across 120 debates, we examine
8 information quantity, quality, and uptake. Within experimental conditions, net initial
9 information distinguishes correct from incorrect debates with an AUC of 0.862,
10 compared with 0.901 for the presence of a correct initial solution. False-claim resistance
11 also discriminates outcomes (0.860), while selective uptake remains inconclusive
12 (0.787, 95% CI [0.474, 0.984]) and the number of distinct initial propositions
13 is less informative (0.532). These findings associate collective success with
14 the quality of initial information and resistance to the spread of false propositions.

15 1 Introduction

16 Multi-agent debate (MAD) is an agentic paradigm in which models exchange reasoning and revise
17 their answers. Debate can improve accuracy over individual responses [Du et al., 2024, Chen et al.,
18 2024], although gains over simpler aggregation methods are inconsistent [Smit et al., 2024, Choi
19 et al., 2025]. Controlled Knight–Knave–Spy (KKS) experiments identify model strength and team
20 diversity as important factors in debate success, and associate externally assessed reasoning quality
21 with successful correction [Wu et al., 2025].

22 Research on human group decision-making shows that possessing useful information does not ensure
23 that groups pool it effectively: discussion can favor shared information over information held
24 by individual members [Stasser and Titus, 1985]. This motivates distinguishing information availability
25 from its subsequent expression in debate, without assuming identical mechanisms in humans
26 and models. Information content also differs from truth: a proposition can exclude many possibilities
27 while being false, making semantic specificity distinct from correctness [Bar-Hillel and Carnap,
28 1953]. Fine-grained factuality evaluation further demonstrates the value of assessing individual
29 claims within responses that mix supported and unsupported content [Min et al., 2023]. We ask:
30 *Which properties of expressed information distinguish successful debates—how much is introduced,
31 how accurate it is, or how selectively it spreads?*

32 We present an analysis framework that translates agent explanations into temporally ordered claims,
33 explicit reasoning relations, and attributed stances. The controlled KKS domain allows us to computationally
34 evaluate translated claim truth and local logical support. Information-theoretic measures
35 characterize how claims narrow possible role assignments, while tracking propositions across agents
36 reveals their subsequent expression.

Our results distinguish information quantity, quality, and uptake. Within experimental conditions, initial information quality and false-claim resistance separate correct from incorrect debates, while proposition count shows little outcome discrimination. Stronger teams re-express true peer claims far more often than false ones. Weaker teams take up both at closer rates. These measurements reveal distinctions that final accuracy and agreement alone cannot resolve, without establishing causal influence between agents.

2 Method

2.1 Experimental setup

We build on the Knight–Knave–Spy (KKS) benchmark and debate protocol of Wu et al. [2025], using independently generated puzzles and an independent implementation. Knights make true statements, knaves make false statements, and spies may do either. Each puzzle has exactly one spy and an exhaustively verified unique solution. Three agents independently propose role assignments, debate each player’s role in sequence, and revise their assignments. After one pass, final assignments are aggregated by role-level majority vote, with a supervisor resolving ties.

We cross team strength (strong, weak) with composition (heterogeneous, homogeneous) on 20 shared six-player puzzles, giving 80 debates. The strong homogeneous team uses three Mercury 2 agents. For AUC, we add 40 fresh eight-player debates to examine outcome associations on larger puzzles. These use the strong heterogeneous roster and a homogeneous team of three GPT-OSS 20B agents on 20 shared puzzles. The strength comparison remains limited to the six-player subset, which includes both strong and weak teams. Strength tiers come from prior solo qualification on the same benchmark pool, not held-out model selection. Configurations and prompts are found in Appendix A and H, respectively.

2.2 Formal-logic translation

Two passes of GPT-5.6 Luna with low reasoning effort translate explanations into claims and explicit reasoning relations. The first extracts propositions in source order, each with a formal expression, exact source span, and commitment tag distinguishing assertions, hypothetical claims, and attributed reports, endorsements, or rejections. Unrepresentable content receives an abstention marker. The second connects the extracted claims through inference, contradiction, and reductio without adding or modifying claims. We preserve invalid or incomplete reasoning without supplying unstated premises (Table 1). Appendix C gives the full specification.

On a ten-field development sample, claim precision was 94.7% and full claim recall was 76.5%. Targeted source reviews identified and corrected additional errors in the analyzed corpus. The reported results use these corrected translations (Appendices D and G).

Table 1: **Example two-pass translation.** The first pass extracts the claims in a sentence. The second identifies how those claims are connected by the stated reasoning.

ID	Formal record	Verbatim source span
Input: “E’s disjunction is false when D is the spy, making E a knave.”		
<i>Pass 1: claim extraction</i>		
c_1	Hyp. (h_1): $\neg \text{STMT_TRUE}(E)$	“E’s disjunction is false”
c_2	Assume (h_1): $\text{ROLE}(D, \text{spy})$	“D is the spy”
c_3	Hyp. (h_1): $\text{ROLE}(E, \text{knave})$	“making E a knave”
<i>Pass 2: relation extraction</i>		
e_1	Infer: $\{c_1, c_2\} \rightarrow c_3$	Full input sentence

2.3 Process-level measures

We measure whether claims are true, how many possible answers they rule out, and whether other agents later repeat them. We check assertions, endorsements, and self-attributed reports against the verified solution. Separately, we check whether the stated premises support their conclusions under

74 the applicable case assumptions and local KKS constraints. This distinguishes a true conclusion
 75 from a valid argument for it.

76 Let Ω contain all assignments with exactly one spy, and Ω_ϕ those satisfying proposition ϕ . For
 77 satisfiable propositions, information content is

$$I(\phi) = \log_2 \frac{|\Omega|}{|\Omega_\phi|}.$$

78 Truth-aligned claims rule out possible assignments while retaining the correct one. Misleading
 79 claims rule out the correct assignment. Redundant claims eliminate no assignments, and contra-
 80 dictory claims leave none. For example, incorrectly claiming that D is the spy rules out five-sixths
 81 of the six-player assignments, including the correct one. High information content does not imply
 82 truth.

83 To measure what information agents start with, we count each distinct initial assertion or endorsement
 84 once per debate, using exact formal-expression matching. *Net initial information* sums truth-aligned
 85 claim bits minus misleading claim bits. Claims can overlap in what they rule out, so their summed
 86 bits need not represent unique information. Uptake tracks whether agents later express their peers’
 87 initial claims. Appendix B gives exact matching rules, denominators, and exclusions.

88 3 Results

89 The corrected six-player corpus contains 3,545 reasoning fields and 34,976 claims, including 24,188
 90 committed claims. Strong teams solved 35 of 40 puzzles and weak teams three of 40, so pooled
 91 outcome comparisons also reflect team strength.

92 3.1 Claim Quality

93 Successful debates had a mean committed-claim accuracy of 94.8%, compared with 50.5% in un-
 94 successful debates. Their mean misleading-claim rates were 5.1% and 48.5%, respectively. Claims
 95 were more often true in successful debates, but most successful debates also came from strong teams.
 96 This comparison cannot tell whether the difference is explained by team strength.

97 3.2 True-Claim Uptake and False-Claim Resistance

98 Eligible uptake pairs link an informative initial proposition to a recipient who did not initially express
 99 it. We exclude explicit role-assignment atoms and match formal expression strings. True-claim
 100 uptake is the fraction of eligible true pairs with a later recipient expression before the final decision.
 101 False-claim resistance is the fraction of eligible false pairs without one. Resistance measures non-
 102 adoption, not explicit rejection.

103 Pooling pairs across compositions within each six-player strength tier, strong teams exhibited higher
 104 true-claim uptake and false-claim resistance (Table 2). Strong teams re-expressed true peer propo-
 105 sitions much more frequently than false ones (81.3% versus 13.4%). The corresponding weak-team
 106 rates were closer (56.3% versus 48.4%). The true-minus-false uptake gaps differed by 60.1 percent-
 107 age points (95% puzzle-cluster bootstrap CI [41.6, 79.1]). This difference describes which claims
 108 the studied teams repeated. Later expression may reflect peer influence, independent derivation, or
 109 previously unstated agreement.

Table 2: Non-role claim–recipient uptake in the 80 size-6 debates after targeted corrections. Rates
 pool pairs within tier. Gap intervals jointly resample the 20 puzzles across all four conditions.

Team	True uptake	False resistance	Gap, pp (95% CI)
Weak	56.3% (130/231)	51.6% (99/192)	7.8 [-7.0, 21.7]
Strong	81.3% (157/193)	86.6% (71/82)	67.9 [56.7, 79.2]

110 3.3 Claim-Level Measures Associated with Debate Success

111 Adding the eight-player extension gives 120 debates across six conditions, with 73 correct and 47
 112 incorrect final outcomes. Within-condition area under the ROC curve (AUC) compares successful

and unsuccessful debates only within the same puzzle size and exact team configuration. Missing metric values are excluded. Intervals jointly resample shared puzzles within each size. Conditions with only one outcome contribute no comparisons (Appendix B).

Net initial information and false-claim resistance were associated with success, with within-condition AUCs of 0.862 and 0.860 (Table 3). Uptake selectivity, defined as true uptake minus false uptake, reached 0.787, but its interval included 0.5. True uptake alone was also inconclusive (0.643 [0.470, 0.819]). The presence of a correct initial solution had an AUC of 0.901, showing that success was associated with having a correct answer before interaction. Claim-level measures also capture useful true propositions accompanying an incorrect complete assignment. We have not tested whether these metrics differ in predictive performance or add information beyond initial accuracy. They also use different eligible samples.

The number of distinct initial propositions showed little outcome discrimination (AUC = 0.532). Counting propositions does not tell us whether agents contribute useful information or which claims other agents repeat. Appendix Table 6 reports all metrics, including pooled AUC and later-proposition counts.

Table 3: Within-condition association with final correctness in 120 debates. Intervals jointly resample shared puzzles. Eligibility varies by metric.

Metric	Eligible debates	AUC (95% CI)
At least one correct initial solution	120	0.901 [0.819, 0.962]
Net initial information	119	0.862 [0.727, 0.953]
False-claim resistance	77	0.860 [0.654, 1.000]
True uptake minus false uptake	71	0.787 [0.474, 0.984]
Distinct initial propositions	119	0.532 [0.348, 0.710]

Separating true from false uptake reveals differences that an overall uptake rate would hide. Stronger teams had higher true-claim uptake and lower false-claim uptake, so their higher uptake was not indiscriminate. Even successful debates contained false claims. Linking claims to their source text and place in the debate lets researchers inspect when false claims first appear and which agents later repeat them.

A next step is to hold an agent’s initial proposal fixed and vary the peer claims it receives. Formal logic could then track whether a correct counterexample leads the agent to withdraw a false proposition, whether an unsupported claim is repeated, and whether these changes precede a corrected final answer. Applying this analysis across debate protocols could help identify which forms of interaction support correction. Such tests require controlled interventions beyond the observational comparisons reported here. Appendix J discusses limitations of the present analysis.

4 Conclusion

The main contribution of this work is a method for observing the claims and reasoning that language models communicate during debate. Formalizing these explanations makes it possible to check claim truth and logical support, measure information content, and follow claims across agents. In KKS, initial information quality and false-claim resistance were associated with successful outcomes, while proposition count was less informative. We hope this approach supports a more detailed study of how debate succeeds or fails and informs the design of multi-agent systems that share and correct information.

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A Debate and model configurations

Table 4 reports the saved model settings for the four size-6 conditions and the two fresh size-8 conditions. The paper’s strong tier corresponds to the repository’s `medium-het` and `medium-hom` run identifiers; this is a naming change, not a change of models.

Table 4: Agent configurations. Parentheses give the reasoning-effort setting; a dash means no explicit setting. All debating agents use temperature 0.7.

Condition	Agents
Size-6 strong heterogeneous	Gemini 3.5 Flash-Lite (medium), Mercury 2 (low), GPT-OSS 20B (medium)
Size-6 strong homogeneous	Three Mercury 2 instances (low)
Size-8 heterogeneous	Gemini 3.5 Flash-Lite (medium), Mercury 2 (low), GPT-OSS 20B (medium)
Size-8 homogeneous	Three GPT-OSS 20B instances (medium)
Size-6 weak heterogeneous	GPT-4.1 Mini 2025-04-14 (–), Gemini 3.1 Flash-Lite (–), Llama 4 Maverick (–)
Size-6 weak homogeneous	Three GPT-4.1 Mini 2025-04-14 instances (–)

Mercury 2 uses OpenRouter with Inception pinned as the provider; GPT-OSS 20B uses DeepInfra. Llama 4 Maverick uses OpenRouter with DeepInfra pinned. Provider fallback is disabled for the pinned OpenRouter routes. The supervisor is GPT-5 Nano 2025-08-07 with medium reasoning effort and temperature 1.0; its assignments are used only for players without a strict majority.

Table 5: Shared execution settings from the saved debate configurations.

Parameter	Setting
Agents / repetitions	Three agents; one repetition per puzzle and condition
Debate depth	One pass through all players (six or eight)
Agent order	Cyclic offset CRC32(task ID) mod 3; fixed within a debate
Accuracy visibility	Hidden
Output contract	Structured JSON; one format-repair retry
Explanation budget	Requested maximum of 100 words; agreement/disagreement fields share the budget
Within-round context	Earlier arguments in the current round are visible
Revision context	Shared pre-revision snapshot; revisions do not see one another
Final submissions	Other agents’ final votes are not visible

The 20 shared puzzle identifiers have prefix `kks-v1-n06-` and suffixes 001, 017, 032, 048, 064, 080, 095, 112, 127, 143, 158, 174, 189, 198, 221, 237, 253, 269, 284, and 300.

The 20 additional size-8 puzzle identifiers have prefix `kks-v1-n08-` and suffixes 001, 017, 032, 048, 064, 080, 095, 111, 127, 143, 158, 174, 190, 207, 221, 237, 253, 269, 284, and 300. These are the fresh matched heterogeneous and GPT-OSS homogeneous runs, not the historical Mercury homogeneous size-8 comparison.

B Detailed metric definitions

B.1 Assignment space and information content

Let w assign one of knight, knave, or spy to each of n players, and let w^* be the verified solution. The information baseline is

$$\Omega = \left\{ w : \sum_{j=1}^n \mathbf{1}[w_j = \text{spy}] = 1 \right\}, \quad |\Omega| = n2^{n-1}. \quad (1)$$

Thus $|\Omega| = 192$ for six-player puzzles and 1,024 for eight-player puzzles. For a proposition ϕ , define

$$\Omega_\phi = \{w \in \Omega : w \models \phi\}, \quad I(\phi) = \log_2 \frac{|\Omega|}{|\Omega_\phi|}, \quad |\Omega_\phi| > 0. \quad (2)$$

The baseline does not impose every puzzle statement as a constraint. A STMT_TRUE expression is evaluated using the referenced statement's canonical formula.

$$\text{category}(\phi) = \begin{cases} \text{contradictory,} & |\Omega_\phi| = 0, \\ \text{redundant,} & |\Omega_\phi| = |\Omega|, \\ \text{truth-aligned,} & 0 < |\Omega_\phi| < |\Omega|, w^* \models \phi, \\ \text{misleading,} & 0 < |\Omega_\phi| < |\Omega|, w^* \not\models \phi. \end{cases} \quad (3)$$

Contradictory expressions receive no finite bit score. Redundancy here means zero reduction from the fixed baseline, not redundancy relative to earlier claims. Summed claim scores need not measure unique information.

Which claims are committed. The results scorer treats a claim as committed when its commitment tag is ASSERT, when it is ENDORSE, or when it is REPORT whose attributed agents include the speaker itself. Before the targeted semantic corrections, the original size-6 corpus contained 35,000 extracted claims, of which 24,221 are committed: 23,313 assertions, 23 endorsements, and 885 self-inclusive reports. Assumptions (612), hypothetical consequences (3,285), opaque records (1,637), reports attributed only to other speakers (5,232), and rejections (13) are not committed. A rejection record stores the rejected proposition, and the record's `effective_expression` field stores its negation, but the scorer does not convert a rejection into a commitment to that negation: rejections carry no committed-claim score and enter no committed-claim rate. The uptake analysis is stricter still and admits only ASSERT and ENDORSE; self-inclusive reports appear there only in a separately labeled sensitivity check (Appendix G).

Debate-level claim-quality statistics. Committed-claim accuracy for a debate is the share of its committed claims with a defined truth value that hold in the verified solution. The truth-aligned, misleading, redundant, and contradictory claim rates are shares of its *scored* committed claims, that is, committed claims assigned one of those four categories; a small number of committed claims are unscored and are excluded from the denominators of the rates but not from the accuracy statistic. Both statistics are computed per debate and then averaged unweighted across debates within an outcome group, so long and short debates count equally.

Initial supply aggregates. Initial-supply measures use scored assertions and endorsements (excluding self-inclusive reports and rejections) whose stage index is the initial-proposal stage, deduplicated within a debate by `effective_expression` where present and `expression` otherwise. Deduplication is by exact string equality of the formal expression; logical equivalence beyond expression equality is not inferred for these measures. Each surviving proposition contributes once regardless of how many agents supplied it, so a proposition present in two or three initial proposals is one unit of supply. Let T and M be the sets of deduplicated truth-aligned and misleading initial propositions. Then

$$\text{distinct initial propositions} = |T| + |M| + |R| + |C|, \quad (4)$$

$$\text{initial claim precision} = \frac{|T|}{|T| + |M|}, \quad (5)$$

$$\text{truth-aligned initial bits} = \sum_{\phi \in T} I(\phi), \quad (6)$$

$$\text{net initial information} = \sum_{\phi \in T} I(\phi) - \sum_{\phi \in M} I(\phi), \quad (7)$$

where R and C are the deduplicated redundant and contradictory propositions. Redundant propositions contribute $I(\phi) = 0$ and contradictory propositions receive no finite bit score, so neither enters the bit sums; both are excluded from initial claim precision, which is undefined when $|T| + |M| = 0$. Bits are summed over overlapping propositions without a union correction, so these are supply totals rather than unique information.

237 B.2 Local inference support

238 Let J contain the players whose statement truth is referenced in a relation, and let $S_j(w)$ evaluate
239 player j 's statement. The local checking universe is

$$\Omega_J = \{w \in \Omega : \forall j \in J, \quad w_j = \text{spy} \vee ((w_j = \text{knight}) \Leftrightarrow S_j(w))\}. \quad (8)$$

240 For the stated premises P and active case assumptions H , let

$$W_{P,H} = \{w \in \Omega_J : w \models \bigwedge P \wedge \bigwedge H\}. \quad (9)$$

241 An inference to ψ is supported when $W_{P,H} \neq \emptyset$ and every world in $W_{P,H}$ satisfies ψ . It is contra-
242 dicted when every such world satisfies $\neg\psi$, and contingent otherwise. Empty $W_{P,H}$ is reported as
243 inconsistent premises. A contradiction relation requires an empty satisfying set. Reductio addition-
244 ally requires a rejected assumption case and a conclusion equivalent to the negation of its assumption.
245 Neutral reports and rejection records are not accepted as truth-bearing premises by this local checker.

246 B.3 Uptake denominators and aggregation

247 The uptake unit is a claim–recipient pair. Within a debate, claims tagged ASSERT or
248 ENDORSE are grouped by formal expression; the primary matching rule is exact equality of
249 `effective_expression` where present and `expression` otherwise. Expressions containing an ex-
250 plicit `ROLE` atom are removed, and a proposition is eligible only if it first appears at the initial-proposal
251 stage and is classified truth-aligned or misleading there; redundant and contradictory propositions are
252 excluded. Let A_ϕ be the agents that introduced ϕ initially. Every one of the three debaters outside
253 A_ϕ contributes one eligible pair, so a proposition introduced by one agent yields two pairs and one
254 introduced by two agents yields one. For pair (ϕ, r) , let $u_{\phi,r} = 1$ if r asserts or endorses ϕ at any
255 stage strictly after the initial proposals and strictly before the final decision, and 0 otherwise. Re-
256 peated re-expression within a pair contributes at most one success, and non-adoption is not coded as
257 explicit rejection.

258 Let $E_{d,T}$ and $E_{d,F}$ be the eligible true and false pairs for debate d . Debate-level rates are

$$A_T(d) = \frac{\sum_{e \in E_{d,T}} u_e}{|E_{d,T}|}, \quad U_F(d) = \frac{\sum_{e \in E_{d,F}} u_e}{|E_{d,F}|}, \quad (10)$$

$$R_F(d) = 1 - U_F(d), \quad S(d) = A_T(d) - U_F(d). \quad (11)$$

259 A zero denominator leaves the rate undefined rather than zero. This is why the eligible samples in
260 Table 3 differ: in the corrected 120-debate cohort, 96 contain an eligible true pair, 77 an eligible false
261 pair, and 71 both. Net initial information is defined for 119 debates, initial precision for 118, and the
262 presence of an exact initial solution for all 120.

263 Tier-level rates in Table 2 are pooled ratios over pairs, not means of debate-level rates:

$$A_T(D) = \frac{\sum_{d \in D} \sum_{e \in E_{d,T}} u_e}{\sum_{d \in D} |E_{d,T}|}. \quad (12)$$

264 This weights debates by their eligible opportunities. It is neither the unweighted mean of debate rates
265 nor the fraction of propositions adopted by at least one peer. Explicit attribution is not required for
266 uptake.

267 B.4 AUC, weighting, and interval construction

268 For an eligible set of debates D , let D_+ and D_- contain correct and incorrect final outcomes, respec-
269 tively, and let x_d be the metric value. Define

$$\text{AUC}(D; x) = \frac{1}{|D_+||D_-|} \sum_{i \in D_+} \sum_{j \in D_-} (\mathbf{1}[x_i > x_j] + \frac{1}{2} \mathbf{1}[x_i = x_j]). \quad (13)$$

270 The within-condition statistic forms this comparison separately inside each of the six
271 size/configuration conditions and then sums concordant pairs and pair counts across conditions be-
272 fore dividing. It therefore weights each condition by its number of eligible correct–incorrect pairs,

and is not an unweighted mean of condition-level AUCs. A condition with no correct or no incorrect eligible outcome contributes no pairs. Debates with a missing value for a given metric are dropped for that metric only, so the eligible sample is metric-specific. This is an outcome-discrimination statistic; it differs from integrating accuracy or agreement over debate rounds, and it is not held-out prediction.

Intervals. For the 120-debate AUC analysis we resample 20 puzzle identifiers with replacement independently within each puzzle size. All condition-level debates associated with a sampled puzzle travel together. This preserves the four matched size-6 conditions and the two matched fresh size-8 conditions. We use 10,000 percentile draws with seed 20260906. A condition with only one outcome contributes no within-condition pairs; a draw is undefined only when the total required denominator is zero. There are 9,980–10,000 defined within-condition AUC draws depending on the metric. Table 2 and the tier-gap contrast use the same size-6 puzzle draws, pooling pair numerators and denominators within each tier; all 10,000 gap draws are defined. These intervals are exploratory and conditional on fixed rosters and translated claims, and do not quantify translation uncertainty. The point estimates reproduce the corrected 120-debate analysis exactly; intervals differ from its earlier condition-independent bootstrap.

The frozen inputs, estimator, bootstrap seed, defined-draw counts, and both pooled and within-condition estimates accompany this merged draft in its revision-support files. All AUCs quoted in the revised main text use this same puzzle-cluster procedure.

B.5 Complete outcome-association results

Table 6: Association with final correctness in 120 size-6 and fresh size-8 debates. Uptake uses non-role proposition–recipient pairs. Intervals jointly resample shared puzzles within each size; missing values are excluded separately for each metric.

Metric	Eligible debates	Pooled AUC	Within-condition AUC (95% CI)
At least one correct initial solution	120	0.870	0.901 [0.819, 0.962]
Net initial information	119	0.823	0.862 [0.727, 0.953]
False-claim resistance	77	0.812	0.860 [0.654, 1.000]
Initial claim precision	118	0.800	0.850 [0.727, 0.943]
Truth-aligned initial bits	119	0.733	0.802 [0.637, 0.921]
Correct distinct initial propositions	120	0.665	0.800 [0.651, 0.912]
True uptake minus false uptake	71	0.855	0.787 [0.474, 0.984]
True-claim uptake	96	0.724	0.643 [0.470, 0.819]
Distinct initial propositions	119	0.462	0.532 [0.348, 0.710]
Distinct later propositions, including final decisions	120	0.734	0.272 [0.136, 0.448]
Distinct later propositions, excluding final decisions	120	0.727	0.258 [0.123, 0.432]

Successful debates tended to introduce fewer distinct propositions later within the same condition, including when final decisions were excluded (AUC = 0.258). This descriptive association does not establish that introducing additional propositions is harmful.

C Translation specification and prompt design

C.1 Record grammar

The first-pass record is ID :: COMMITMENT :: EXPRESSION :: SPAN. Table 7 gives the commitment inventory. Case records use hN :: CASE(STATUS) :: SPAN, with status ACCEPTED, REJECTED, or UNRESOLVED. A case outcome is omitted when unstated.

The expression vocabulary includes ROLE, STMT_TRUE, SAME_ROLE, NOT, AND, OR, XOR, IFF, IMPLIES, COUNT, and PARITY. ABSTAIN can also preserve an unresolved assumption, hypothetical, or report. Arbitrary quantifiers, nested proof structures, and unstated premises are outside the representation. Exact spellings, arguments, and extraction examples appear in the full prompt below.

Table 7: Commitment tags in the stable claim prompt. The field metadata identifies the speaker; attributed targets identify the source of the reported proposition.

Tag	Meaning
ASSERT	Unconditional assertion
ASSUME(hN)	Assumption opening a hypothetical case
HYPOTHETICAL(hN)	Consequence inside the named case
REPORT(targets)	Attributed proposition without endorsement
ENDORSE(target)	Acceptance of an attributed proposition
REJECT(target)	Rejection of the recorded proposition
OPAUQUE	Meaningful content represented as ABSTAIN

305 C.2 Model calls and constraints

306 Both semantic passes use GPT-5.6 Luna with low reasoning effort through the Codex CLI. Saved
 307 records identify claim prompt v5, relation prompt v1, and format kks-minimal-relations-v2.
 308 The first-pass prompt receives puzzle statements and canonical expressions, field metadata, agent
 309 IDs, submitted roles, and source text. Submitted roles are context only. Its terminal override restricts
 310 output to claims and cases; any first-pass relation lines are discarded. The second pass receives the
 311 original source and fixed claim/case inventory.

312 The prompt requires verbatim spans, preserves negation and alternatives, and separates statement
 313 content from evaluations of its truth. It forbids adding premises to make an argument valid. Relation
 314 outputs use only INFER, CONTRADICT, or REDUCTIO records referencing fixed identifiers. When no
 315 relation is extracted, the output is NONE.

316 Optional Luna-high format repair is restricted to parser failures. A repair must reduce errors while
 317 preserving already-valid records. Subsequent cleanup uses separately saved Sol-medium repair out-
 318 puts and explicit correction ledgers; original responses remain unchanged. No parse-clean status is
 319 interpreted as semantic correctness.

320 D Supplementary corpus and validation tables

Table 8: Original size-6 inventory after parser cleanup, before the later semantic correction overlay. Each row contains the same 20 puzzles. Claim counts are occurrences, not distinct propositions. Case counts are explicit case-outcome records.

Condition	Fields	Claims	Cases	Relations
Strong heterogeneous	880	8,489	136	3,194
Strong homogeneous	883	8,548	98	3,490
Weak heterogeneous	909	8,461	121	3,111
Weak homogeneous	873	9,502	177	3,324
Total	3,545	35,000	532	13,119

321 Each condition contributes 60 initial explanations, 360 revision explanations, and 60 final explana-
 322 tions. The remaining fields are agreement/disagreement reasoning: 400, 403, 429, and 393, respec-
 323 tively. A debate argument can contribute separate agreement and disagreement fields; fields are not
 324 synonymous with turns.

325 Expression recall excludes unrepresentable reference claims; full claim recall additionally assesses
 326 commitment and attribution. Case membership is scored separately. Strict relation matching requires
 327 the reviewed endpoints and structure. This development run used no model-based format repair and
 328 retained two claim-stage parser errors. The 40/40 verdict score is conditional on matching a relation,
 329 not end-to-end accuracy.

330 **Ambiguity and reference annotations.** Natural-language explanations can admit multiple plau-
 331 sible formalizations, particularly in hypothetical scope, attribution, and implicit premises. Develop-

Table 9: Original size-6 relation types and residual parser errors before and after parser cleanup, preceding the semantic correction overlay. Pre-cleanup errors are those remaining in the saved generation artifact, after any generation-time repair. Error counts are not counts of semantic mistakes.

Condition	Inference	Contradiction	Reductio	Before	After
Strong heterogeneous	3,112	65	17	84	0
Strong homogeneous	3,447	43	0	76	0
Weak heterogeneous	3,014	93	4	42	0
Weak homogeneous	3,207	104	13	111	0

Table 10: Two-pass translator development evaluation on ten four-/five-player reasoning fields, with 136 reference claims, 61 relations, and four cases. This development sample is separate from the analyzed six-player corpus and is not a held-out validation set.

Measure	Matched / eligible	Percentage
Expression recall	101/118	85.6%
Full claim recall	104/136	76.5%
Claim precision	107/113	94.7%
Commitment/attribution accuracy on matched expressions	98/101	97.0%
Case-member recall	26/37	70.3%
Strict relation recall	40/61	65.6%
Strict relation precision	40/46	87.0%
Local-verdict accuracy on strict relation matches	40/40	100.0%

ment scores therefore measure agreement with the reviewed reference annotations, which may not exhaust all defensible interpretations. Reference annotations were reviewed with model assistance. Disagreements can reflect either a translation error or an alternative formalization of ambiguous text.

E Code availability and computational cost

Code and data availability. The accompanying anonymized supplementary archive contains the 120 raw debate outputs across all six conditions, benchmark questions and verified solutions, exact configurations and prompts, all six translation sources, the source-level correction ledger and overlays, scored data, analysis code, and puzzle-cluster bootstrap code. Its README.md supplies dependency versions and commands; MANIFEST.json records file hashes, and COMPUTE.md documents resource accounting and its limits. The complete source-to-table replay was verified from the extracted package, including correction integration, source-level rescoring, metric regeneration, and the 10,000-draw puzzle-cluster intervals. Reproducing the reported tables from these archived outputs requires no new model calls. Earlier size-6 sensitivity analyses are separately labeled and are not substituted for the current corrected estimates.

Computational cost. The original 80 size-6 debates incurred \$7.54 in recorded API costs across 3,372 completed calls, including logged format-repair calls. Six additional failed calls have no recorded cost. These log-based amounts have not been reconciled against provider invoices.

Table 11: Recorded usage for the original 80-debate size-6 corpus; excludes the fresh size-8 extension and subsequent semantic review. Input includes cached tokens; output includes reasoning tokens. Translation and cleanup used Codex CLI, for which no per-call monetary cost is recorded.

Component	Input tokens	Output tokens	Recorded USD
Debate generation	22,328,302	1,872,915	\$7.54
Translation and format repair	97,956,899	5,467,118	Unavailable
Sol cleanup (two passes)	7,372,825	224,286	Unavailable

Translation usage covers 7,761 returned model calls, including generation-time format repair. Sol cleanup covers 305 returned calls, including rejected repairs. The combined Codex CLI input total

includes 71,188,224 cached tokens. These figures exclude qualification runs, other puzzle sizes, development experiments, and additional review not captured in this accounting; they do not represent the total research cost. Failed calls without usage records are not assigned zero cost.

API-equivalent estimate. Applying published standard-context API prices to the recorded CLI tokens gives \$14.21 for Luna translation and format repair and \$16.45 for Sol cleanup. Together with the recorded debate subtotal, this yields \$38.21; without input-cache discounts, the corresponding estimate is \$67.67. Rates per million input/cached-input/output tokens are \$0.20/\$0.02/\$1.20 for Luna and \$4.00/\$0.40/\$20.00 for Sol, checked on September 5, 2026.¹ These are API-equivalent scenarios, not subscription expenditure. They assume standard-context pricing and, in the first scenario, the recorded cache-hit counts. Per-request sizes cannot be recovered from the aggregate CLI records, so possible long-context surcharges are excluded; these figures are not billing bounds.

Compute resources. All model inference ran on hosted provider endpoints; their accelerator and memory configurations are not disclosed. The original weak conditions and size-6 translation pipeline used an Apple M1 workstation with 8 cores, 8 GB memory, and macOS 26.3. The original strong conditions and fresh size-8 debates were orchestrated on standard GitHub-hosted ubuntu-latest runners, not a second personal workstation. Their setup logs identify Ubuntu 24.04 images 20260720.247.2 and 20260831.293.1, respectively. GitHub documents this private-repository Linux runner class as x64 with 2 virtual CPUs and 8 GB RAM; these are class specifications rather than measurements from the individual jobs.² The corrected 120-debate analysis and puzzle-cluster reconciliation ran locally on an Apple M5 Pro with 18 CPU cores, 24 GB memory, and macOS 26.5.1. No local accelerator was used for model inference.

The fresh size-8 extension records 2,207 model calls, 25,177,180 tokens, and \$3.52 in recorded API cost across both conditions. These are saved usage totals, not reconciled invoices; the supplementary COMPUTE.md identifies their raw sources.

The fresh size-8 job’s debate-execution step lasted 72 minutes 17 seconds, including retries; this is elapsed time for the matched two-condition run, not the sum of individual debate durations. The archived original strong results include recovery checkpoints, so the final checkpoint job duration is not the duration of original generation. Complete token and elapsed-time accounting for the fresh size-8 translation and subsequent semantic review is not consolidated; the tables below and above remain explicitly scoped to the original size-6 study.

The work is bound by provider latency rather than local computation; the original size-6 analysis scripts hold the corpus in memory as data frames and complete in minutes on the M1. The saved debate configurations set a global client concurrency of 80 in-flight requests, with per-provider caps of 40 for OpenAI, 40 for Gemini, and 8–20 for OpenRouter routes; recovery passes used a concurrency of 3.

Table 12 reports per-debate wall-clock time for the analyzed corpus, recovered from the run_started and run_completed events of each debate. Summing across all 80 debates gives 11.6 hours of debate execution. Because the analyzed six-player puzzles were run concurrently with other puzzle sizes inside the same experiments, this total is the serial cost of the reported debates rather than the elapsed duration of any experiment. Translation and cleanup were issued through the Codex CLI, which records a creation timestamp per translated field but no per-call duration; those records span two calendar days with idle gaps between sessions, so we do not report a translation wall-clock total.

Storage for the original released snapshot is dominated by run event logs and translation records. The token, call-count, and recorded-cost accounting above covers the original size-6 corpus only and excludes qualification runs, other puzzle sizes, development experiments, and failed or superseded attempts, so it understates total project compute.

¹<https://developers.openai.com/api/docs/models/gpt-5.6-luna>; <https://developers.openai.com/api/docs/models/gpt-5.6-sol>.

²<https://docs.github.com/en/actions/reference/runners/github-hosted-runners>

Table 12: Per-debate wall-clock time for the 80 analyzed six-player debates, in seconds. Each condition contains 20 debates. Times include all provider latency, retries, and format repair for that debate.

Condition	Median	Mean	Range
Strong heterogeneous	520	819	300–2,810
Strong homogeneous	226	232	210–262
Weak heterogeneous	864	869	841–901
Weak homogeneous	168	169	152–182

398 F Assets, versions, and licenses

399 Table 13 lists the third-party assets this work depends on. The debate protocol, phase structure,
400 and Knight–Knave–Spy task family follow Wu et al. [2025]; we credit that work as the source of
401 the protocol and generate our own puzzles rather than reusing theirs. The procedurally generated
402 puzzles, all debate transcripts, all translation records, and all analysis code are produced by us and
403 are not derived from a third-party dataset. We release the code under the MIT License and the puzzles,
404 transcripts, translation records, and analysis tables under CC-BY-4.0. Hosted models were accessed
405 under their providers’ terms of service and are not redistributed; only their outputs to our prompts
406 appear in the release.

Table 13: Third-party assets. Hosted models are accessed as services rather than redistributed artifacts, so provider terms of service govern them in place of a software license.

Asset	Role	Terms
GPT-4.1 Mini 2025-04-14, GPT-5 Nano 2025-08-07, GPT-OSS 20B	Debater and supervisor models	Provider terms of service
Gemini 3.1 Flash-Lite, Gemini 3.5 Flash-Lite	Debater models	Provider terms of service
Mercury 2, Llama 4 Maverick	Debater models	Provider terms of service
GPT-5.6 Luna, GPT-5.6 Sol	Translation, format repair, cleanup	Provider terms of service
OpenAI, Google, OpenRouter, DeepInfra, Inception	Model serving routes	Provider terms of service
Codex CLI	Translation call harness	Vendor terms of service
Python 3.11+	Runtime	PSF License
NumPy, pandas	Analysis	BSD 3-Clause
LiteLLM, Pydantic, PyYAML, python-dotenv, Typer	Harness and configuration	MIT
Pillow	Asset rendering	MIT-CMU

407 G Claim-uptake robustness and attribution audit

408 **Version boundary.** This section preserves the prior size-6 robustness study and its four-polarity
409 correction overlay from the incoming manuscript. Its tables are historical sensitivity results and were
410 not regenerated under the broader 75-operation correction overlay used in the revised main text. They
411 must not be substituted for the current main-table estimates. The current overlay and its review scope
412 are documented immediately below; the older study remains available for inspection rather than being
413 silently combined with it.

414 **Current targeted semantic corrections.** A bounded assistant diagnostic audit inspected eight ini-
415 tial explanations (139 extracted records) and 29 uptake opportunities across 20 size-6 debates. Three
416 Sol review assignments and an independent polarity review then checked targeted errors and recur-
417 ring patterns in the six source corpora. The correction pass changed 24 fields across 22 debates:
418 25 claim replacements, 48 removals, and two additions. One explicit reasoning relation was added;
419 relation endpoints and all unaffected fields were checked. The original 5,826-field inventory was
420 screened for two narrow error patterns; 43 further generic-consistency candidates remain unreviewed,

and ambiguous scope decisions were left unchanged. The review addressed the selected fields and recurring error patterns.

The corrected size-6 corpus contains 34,976 claims and 24,188 committed claims; the fresh size-8 corpus contains 25,232 claims. Raw inputs remain unchanged. Source-level regeneration yields 968 eligible uptake pairs across 120 debates, versus 990 before correction. The current size-6 strong-minus-weak uptake-gap contrast is 60.1 percentage points [41.6, 79.1]. Exact-expression matching, non-role eligibility, commitment policy, and stage boundaries are unchanged. Differences from the older tables below reflect the correction scope, not changes in final outcomes.

Rates pool eligible pairs within tier; repeated occurrences count once per pair. The primary matching rule is exact equality of effective formal expressions, not unrestricted semantic equivalence. There are 20 independent puzzle clusters, four conditions per puzzle, and one run per condition–puzzle. The bootstrap samples 20 puzzle identifiers with replacement jointly across all conditions, retaining every claim, recipient, and truth category inside each sampled cluster. We use 10,000 percentile draws (seed 20260906). Zero denominators remain undefined; the released tables record the number of defined draws. These are exploratory intervals conditional on fixed rosters and translated claims, not model-population or translation-uncertainty intervals. The primary inferential contrast is the between-tier difference in the true-minus-false uptake gap; the checks below are descriptive, without claims of multiplicity-adjusted significance.

Table 14: Composition-specific uptake counts and gaps. Each condition contains 20 matched debates. Gaps are percentage points.

Composition / tier	True uptake	False resistance	Gap (95% CI)
Heterogeneous / strong	80/98	50/54	74.2 [62.7, 85.1]
Heterogeneous / weak	64/112	49/89	12.2 [-5.3, 29.6]
Homogeneous / strong	79/97	27/36	56.4 [36.0, 80.5]
Homogeneous / weak	66/119	50/103	4.0 [-16.1, 23.0]

Table 15: Exploratory sensitivity analyses. Values are strong-minus-weak differences in the true-minus-false uptake gap, in percentage points.

Restriction	Gap difference (95% CI)
Semantic matching (original corpus)	62.9 [43.3, 82.7]
Include self-reports (original corpus)	58.7 [41.4, 76.3]
Corrected primary	59.3 [42.8, 76.2]
Initially incompatible claims	68.7 [56.2, 81.7]
Uptake in argument fields only	23.2 [7.7, 40.1]
Uptake before last round	58.1 [41.1, 75.7]
Shared claims, equal stratum weights	79.4 [40.4, 108.9]

The shared-claim check retains puzzle/composition/expression strata represented in both tiers, averages uptake across recipients within each stratum, and weights shared strata equally; only 9,999 of 10,000 gap draws are defined. Semantic matching compares proposition truth signatures over all 192 exactly-one-spy role worlds per puzzle, not merely the verified solution. Initially incompatible means false under the recipient’s submitted initial role assignment. Argument-only and early checks restrict which uptake events count while retaining the primary eligibility denominator. Self-inclusive reports include the speaker among the attributed agents. The semantic and self-report sensitivities use the original corpus; the remaining checks use the disclosed correction overlay. None of these restrictions randomizes claim supply.

Attribution audit. Among the 708 original eligible pairs, no uptake event was tagged as an explicit ENDORSE of an initial introducer. A broader proxy required a reference to an initial introducer plus a separate commitment to the same proposition in the same reasoning field: it identified three weak-team pairs (two true, one false) and zero strong-team pairs. The audit artifact contains all 78 matching reference occurrences, including non-endorsements. Including self-inclusive reports as commitments yields 5 true and 0 false attributed pairs for strong teams, versus 23 true and 26 false for weak teams; the between-tier gap contrast is 6.1 points [-3.0, 15.7]. These broader occurrences

455 often say only that agents agree; they do not establish that the speaker changed because of a peer.
456 A source-text spot-check also found explicit peer credit missing from the attribution tags. Annota-
457 tion sparsity and incomplete attribution capture therefore preclude a reliable influence comparison.
458 Zero observed events do not establish a zero population rate, and a degenerate bootstrap interval is
459 uninformative.

460 **Reviewed sample and polarity corrections.** An assistant reviewed full origin and first-uptake
461 fields for 16 reproducibly sampled adopted pairs, two per condition and truth category. Fourteen
462 were supported, one had an inverted truth-value translation, and one had an ambiguous conditional
463 origin. A corpus-wide conservative full-span polarity screen then found four clear mismatches; all
464 four full source fields were reviewed and corrected in a separate analysis overlay. This changes strong
465 false uptake from 14/90 to 13/90 and leaves the other pooled counts unchanged. The original trans-
466 lation artifacts remain intact. This targeted review is not an independent human validation study
467 and does not estimate coverage or the error rate among non-adoptions. The ambiguous pair remains
468 disclosed rather than silently relabeled. The corpus-wide claim-quality statistics in Section 3.1 and
469 the reasoning-relation counts in Table 9 were not regenerated from this four-claim overlay and are
470 reported as original-corpus results.

471 H Full debate prompts

472 The shared system instruction is combined with the phase-specific user template. Braced variables
473 are populated at runtime; doubled braces in the templates escape literal JSON braces. These tem-
474 plates implement the phase structure adapted from Wu et al. [2025].

475 H.1 Shared system instruction

There are {size} players in the game, each assigned a role of knight, knave, or spy. Each player will make a statement about themselves and other players.

Game Rules:

- Knights always tell the truth.
- Knaves always lie.
- Spies can either tell the truth or lie.

Your task is to deduce the role of each player based on the statements.

You are {agent_id}, serving as the {role} in a multi-agent debate.
{role_description}

CRITICAL REQUIREMENTS:

{assignment_requirements}

- Base your decision on the game logic and evidence from the statements
- If you're uncertain, choose the most likely role based on available evidence

Please follow strictly the format of the return, or the response will be rejected.

476 H.2 Initial proposal

{task}

This is your initial proposal. Solve the game independently before seeing any other agent's work.

For this run, use the following return format instead of the return format above:

```
{{  
  "roles": {{  
    "player_name": "role"  
  }},  
  "explanation": "string"  
}}
```

The roles object contains one entry for each player, mapping the player's name to the player's role. Explanation is a string that contains the argument to derive the result. Keep your explanation having details but less than 100 words. Return only valid JSON.

477 H.3 Player-level argument

{task}

This is a debate turn about {player}. Review the other agents' role assignments and arguments, while checking them against the complete game.

The context contains the available latest solutions and any previous debate rounds or arguments already made in this turn. If your own latest solution is present, it is under {agent_id}.

{context}

Make a definitive role assignment for {player}, then classify the peer agents.

Your agent ID is exactly: {agent_id}
The only peer agent IDs you may classify are: {other_agent_ids}

Copy each peer ID exactly once into either `agree\_with` or `disagree\_with`. Never include your own ID ({agent_id}), a player name, or any other ID. The two arrays must not overlap. Use an empty array and an empty reasoning string when there is nobody in that category.

For this turn, use the following return format:

```
{{
  "player": "{player}",
  "role": "role",
  "agree_with": ["peer_agent_id"],
  "disagree_with": ["peer_agent_id"],
  "agree_reasoning": "string",
  "disagree_reasoning": "string"
}}
```

Keep the two reasoning strings under 100 words combined. Return exactly one JSON object, with no Markdown fences, preamble, commentary, or revised draft.

478 **H.4 Complete-assignment revision**

{task}

The group has just debated {player}. Review the arguments and revise your complete solution for every player, not only {player}.

{context}

For this run, use the following return format instead of the return format above:

```
{{
  "roles": {{
    "player_name": "role"
  }},
  "explanation": "string"
}}
```

The roles object contains one entry for each player, mapping the player's name to the player's role. Explanation is a string that contains the argument to derive the result. Keep your explanation having details but less than 100 words. Return only valid JSON.

479 **H.5 Final decision**

{task}

This is the final discussion phase. Privately reflect on your latest complete solution using all available evidence in the context, then submit your final complete solution independently from the other agents' final votes. When a verifier report is present, treat it only as error-checking evidence and perform your own final logical check before answering.

{context}

For this run, use the following return format instead of the return format above:

```
{{
  "roles": {{
    "player_name": "role"
  }},
  "explanation": "string"
}}
```

The roles object contains one entry for each player, mapping the player's name to the player's role. Explanation is a string that contains the argument to derive the result. Keep your explanation having details but less than 100 words. Return only valid JSON.

480 H.6 Supervisor tie-breaking

```
{task}
```

The debating agents did not reach a strict majority for the players listed in tied_players. Review the complete debate history and produce a logically consistent complete solution. Your assignments will be used only for those tied players.

```
{context}
```

For this run, use the following return format instead of the return format above:

```
{{
  "roles": [{
    "player_name": "role"
  }],
  "explanation": "string"
}}
```

The roles object contains one entry for each player, mapping the player's name to the player's role. Explanation is a string that contains the argument to derive the result. Keep your explanation having details but less than 100 words. Return only valid JSON.

481 I Full translator prompts

482 The first-pass template is reproduced with its terminal claims-only override. Historical relation ex-
483 amples remain in the prompt, but the host discards first-pass relation output. Dollar-prefixed place-
484 holders supply field context or fixed records.

485 I.1 Claim pass: system instruction

Extract explicit atomic commitments and flat hypothetical cases only. Preserve
↪ negation and attribution. Do not output reasoning relations. Every source span
↪ must be copied verbatim.

486 I.2 Claim pass: user template

Task

Translate one KKS reasoning field into atomic claims and explicit local relations.

Puzzle statements (`speaker :: text :: canonical expression`):

\$statements

Rules: each player has exactly one role; exactly one is a spy; knights speak
↪ truthfully; knaves falsely; spies may say either.

Field metadata: \$location

Debate agents: \$debate_agents

Submitted roles (context only; do not add claims that the source does not state):
↪ \$submitted

Source field:

\$source_text

Return one line per explicit proposition, in source order:

`c1 :: COMMITMENT :: EXPRESSION :: exact verbatim source span`

COMMITMENT is exactly one of:

- `ASSERT`: the current speaker presents the proposition as unconditionally true.
- `ASSUME(h1)`: the proposition opens or defines hypothetical case h1.
- `HYPOTHETICAL(h1)`: the proposition is claimed only within case h1.
- `REPORT(player:A,player:B)` or `REPORT(agent:agent-1,agent:agent-2)`: attributed
↪ without endorsement. List every target when the same proposition is reported
↪ for several players or agents. The reporting speaker is already field
↪ metadata. "All agents agree/assign/conclude P" is REPORT of P to every listed
↪ debate agent, never ASSERT. Include the field speaker too when the source says
↪ "all". "All peers" means every listed debate agent except the field speaker.
↪ Its expression may be ABSTAIN only when the reported content is
↪ unrepresentable.
- `ENDORSE(agent:agent-1)`: the current speaker accepts that agent's proposition.
- `REJECT(agent:agent-1)`: encode the proposition being rejected, not its negation.
- `OPAQUE`: a meaningful proposition that cannot be represented; expression ABSTAIN.

ASSUME(hN), HYPOTHETICAL(hN), and REPORT(...) may use ABSTAIN when their
↪ proposition is meaningful but unrepresentable. This retains case or
↪ attribution metadata while marking expression translation as unavailable.

Optionally record each explicit case outcome on a separate line:

`h1 :: CASE(REJECTED|ACCEPTED|UNRESOLVED) :: exact verbatim outcome span`

REJECTED means the source says the case is impossible or fails; ACCEPTED means the
 ↪ source says it works or adopts it; UNRESOLVED means the source explicitly
 ↪ leaves it open or uncertain. Do not infer an outcome from the verified answer.

Flat cases only: reuse one h-ID for all claims inside the same case. If the prose
 ↪ gives no explicit outcome, omit the case line.

Optionally record a local reasoning relation only when the source explicitly
 ↪ connects already translated claims:

`e1 :: INFER :: FROM(c1,c2) :: TO(c3) :: exact full relation span`

`e2 :: CONTRADICT :: FROM(c1,c2) :: exact full relation span`

`e3 :: REDUCTIO :: ASSUME(c1) :: VIA(e2) :: TO(c3) :: exact full relation span`

- INFER means the source presents the FROM claims as reasons for the TO claim.
- CONTRADICT means the source says the listed claims cannot jointly hold.
- REDUCTIO means an ASSUME claim leads through a stated CONTRADICT relation to the
 ↪ negation of that assumption. The TO claim must be exactly that negation.
- Every TO(...) contains exactly one claim ID. If the conclusion is "P or Q",
 ↪ translate one OR(P;Q) claim and point TO it. If the source separately
 ↪ concludes both P and Q, emit two INFER lines, each with one TO claim.
- Do not use REDUCTIO merely to choose another role after one role is rejected.
 ↪ For "C is knave or spy; not knave; therefore spy", use an OR claim, the
 ↪ negated knave claim, and ordinary INFER. REDUCTIO is only
 ↪ assumption-to-exact-negation.
- The relation span is the smallest exact contiguous source span containing all
 ↪ referenced claim spans and the connective language such as because, so, thus,
 ↪ impossible, or contradiction. Do not emit character offsets.
- Do not invent missing premises or connect claims merely because the puzzle makes
 ↪ the inference valid. A malformed or invalid relation does not erase its claims.
- Before emitting a relation, verify that the exact source span of every cited
 ↪ claim occurs inside the relation span. Similar wording elsewhere is not
 ↪ enough. If the source repeats a proposition in new wording, translate that
 ↪ occurrence as a new claim or omit the relation.

Inventory rules:

- "If", "assume", "suppose", and counterfactual "were" clauses that introduce a
 ↪ case use ASSUME(hN), not merely HYPOTHETICAL(hN). Consequences use the same hN.
- ASSUME is only for a genuine case opener. A premise or parenthetical such as
 ↪ "(only one spy)" inside an active case is HYPOTHETICAL in that existing case;
 ↪ it does not open a nested case.
- Restrictive language such as "P only fits if A is knave" asserts the required
 ↪ condition when the prose presents P as established; it does not open a new
 ↪ case.
- A case remains active until the prose explicitly rejects, accepts, or exits it.
 ↪ A period alone does not close a case. Subsequent deductions and summaries stay
 ↪ HYPOTHETICAL(hN) until the source explicitly returns to an unconditional
 ↪ result. After a nested case closes, subsequent provisional claims return to
 ↪ the outer case.
- Every proposition inside an active case inherits that case, including short
 ↪ parentheticals such as "(true)" and conclusions after a nested case closes.
 ↪ The one exception is the fixed content of a puzzle character's statement: keep
 ↪ that as REPORT(player:X), because the character said the same thing in every
 ↪ case. Its truth evaluation and any deductions from it do inherit hN.
- Descriptions such as "the consistent solution where A is knave" or "a solution
 ↪ with D as spy" are assertions about that solution, not hypothetical cases. Do
 ↪ not use HYPOTHETICAL unless the source actually opens an if/assume/suppose
 ↪ case.
- Translate every explicit truth evaluation separately from statement content: "C
 ↪ says P, which is false" contains REPORT(player:C) of P and ASSERT of
 ↪ NOT(STMT_TRUE(C)). Do the same for "which is true".

- Split explicit role lists and conjunctions into all of their atomic claims.
 - ↪ Never omit "A is knave" or "B is knight" because they occur in one sentence.
- Never split alternatives into separate assertions. "C is either knave or spy" is
 - ↪ one OR(ROLE(C,knave);ROLE(C,spy)) claim, not two role claims.
- Preserve explicit parenthetical claims such as "C is knave (not knight)" as both
 - ↪ ROLE(C,knave) and NOT(ROLE(C,knight)).
- Emit explicit rules and premises introduced by "since", "because", or "with". Do
 - ↪ not duplicate an earlier claim merely because it is restated. Create a new
 - ↪ occurrence only when its commitment/case/attribution changes or a relation
 - ↪ needs that exact repeated span as an endpoint.

Expressions:

```
`ROLE(A,knight)`, `STMT_TRUE(D)`, `NOT(expr)`, `SAME_ROLE(A,B)`, `AND(expr;expr)`,
↪ `OR(expr;expr)`, `XOR(expr;expr)`, `IFF(expr;expr)`, `IMPLIES(expr;expr)`,
↪ `COUNT(exactly,1,spy,A,B,C,D)`, `PARITY(even,knave,A,C)`, or `ABSTAIN`.
```

Meaning rules:

- Preserve every negation and its scope literally.
- "D cannot be knight" is NOT(ROLE(D,knight)).
- "B's statement is false" is NOT(STMT_TRUE(B)); it is not the negation of a
 - ↪ convenient fragment of B's statement.
- "Condition (1) is false" negates condition (1), not the speaker's whole statement.
- A puzzle character saying P is REPORT(player:X) of P, not ASSERT of P.
- Use IMPLIES(P;Q) only when the source explicitly asserts the conditional
 - ↪ proposition "if P, then Q". When the source actually reasons "because P, Q",
 - ↪ emit P and Q as claims plus INFER FROM(P) TO(Q), not IMPLIES(P;Q).
- A conclusion after a rejected case is ASSERT only when the source explicitly
 - ↪ adopts it outside the case.
- Copy one exact contiguous source span. Never paraphrase the span.
- Never shorten copied text with `...` or an ellipsis character. If a span is
 - ↪ long, copy the entire exact substring anyway.
- ABSTAIN is never paired with ASSERT, ENDORSE, or REJECT. Use OPAQUE for an
 - ↪ unconditional meaningful proposition that cannot be expressed. Use ABSTAIN
 - ↪ with ASSUME, HYPOTHETICAL, or REPORT only to preserve that metadata.

Example source: `Assume A is knight. Then B is knave. Impossible, so A is not knight.`

```
```text
c1 :: ASSUME(h1) :: ROLE(A,knight) :: Assume A is knight
c2 :: HYPOTHETICAL(h1) :: ROLE(B,knave) :: Then B is knave
h1 :: CASE(REJECTED) :: Impossible
c3 :: ASSERT :: NOT(ROLE(A,knight)) :: so A is not knight
e1 :: CONTRADICT :: FROM(c1,c2) :: Assume A is knight. Then B is knave. Impossible
e2 :: REDUCTIO :: ASSUME(c1) :: VIA(e1) :: TO(c3) :: Assume A is knight. Then B is
↪ knave. Impossible, so A is not knight
```
```

Example source: `Player C says B is spy, which is false, so C is knave.`

```
```text
c1 :: REPORT(player:C) :: ROLE(B,spy) :: Player C says B is spy
c2 :: ASSERT :: NOT(STMT_TRUE(C)) :: which is false
c3 :: ASSERT :: ROLE(C,knave) :: so C is knave
e1 :: INFER :: FROM(c2) :: TO(c3) :: which is false, so C is knave
```
```

Example source: `If A's statement is true, A is knight; if false, A is knave.`

```
```text
c1 :: ASSUME(h1) :: STMT_TRUE(A) :: If A's statement is true
c2 :: HYPOTHETICAL(h1) :: ROLE(A,knight) :: A is knight
c3 :: ASSUME(h2) :: NOT(STMT_TRUE(A)) :: if false
```

```
c4 :: HYPOTHETICAL(h2) :: ROLE(A,knave) :: A is knave
```
```

Example source: `Assume A is knight. If C is knave, contradiction. So C is spy.`

```
```text
c1 :: ASSUME(h1) :: ROLE(A,knight) :: Assume A is knight
c2 :: ASSUME(h2) :: ROLE(C,knave) :: If C is knave
h2 :: CASE(REJECTED) :: contradiction
c3 :: HYPOTHETICAL(h1) :: ROLE(C,spy) :: So C is spy
```
```

Example source: `Agents 1 and 2 say B is knight. B's statement is true because
↪ exactly one spy exists.`

```
```text
c1 :: REPORT(agent:agent-1,agent:agent-2) :: ROLE(B,knight) :: Agents 1 and 2 say
↪ B is knight
c2 :: ASSERT :: STMT_TRUE(B) :: B's statement is true
c3 :: ASSERT :: COUNT(exactly,1,spy,A,B,C,D) :: exactly one spy exists
e1 :: INFER :: FROM(c3) :: TO(c2) :: B's statement is true because exactly one spy
↪ exists
```
```

STAGE 1 OUTPUT OVERRIDE

- Return only cN claim lines and optional hN case lines.
- Do not return any eN relation lines in this pass. A separate call extracts
↪ relations after claims and cases are fixed.
- Do not repeat a claim solely to serve as a relation endpoint.

Return only claim and case lines, without Markdown or explanation.

487 I.3 Relation pass: system instruction

Extract local relation records only. The supplied claims are immutable; do not
↪ reconstruct a complete proof.

488 I.4 Relation pass: user template

Task

Extract only explicitly stated local relations between fixed claims.

Source field:

\$source_text

Fixed parsed claims (read-only):

\$fixed_claims

Fixed parsed cases (read-only):

\$fixed_cases

Return zero or more relation lines in source order, using only these shapes:

`e1 :: INFER :: FROM(c1,c2) :: TO(c3) :: exact full relation span`

`e2 :: CONTRADICT :: FROM(c1,c2) :: exact full relation span`

`e3 :: REDUCTIO :: ASSUME(c1) :: VIA(e2) :: TO(c3) :: exact full relation span`

- You cannot add, edit, renumber, or repeat claims or cases. Output eN lines only.
- Use only claim IDs listed above. Every T0 contains exactly one fixed claim ID.
- INFER means the source presents every FROM claim as a reason for the T0 claim.
- CONTRADICT means the source says the FROM claims cannot jointly hold.
- REDUCTIO is strict: an ASSUME claim leads through a stated CONTRADICT relation
 ↪ to a fixed claim that is exactly the negation of that assumption.
- Do not use REDUCTIO merely to choose a remaining role after rejecting one.
- Do not invent missing premises or edges that are merely logically valid.
- The relation span is the smallest exact contiguous source substring containing
 ↪ every referenced claim span, every referenced relation span, and the
 ↪ connective language. If no such substring exists, omit the relation.
- Do not reconstruct a complete proof or add transitive/unstated edges.

If the source states no relations between the fixed claims, return exactly `NONE`.
 ↪ Otherwise return only relation lines, without Markdown or explanation.

489 **I.5 Claim format repair: system instruction**

Perform format-only repair. Preserve substantive translation choices, including
 ↪ logically questionable relations.

490 **I.6 Claim format repair: user template**

Format-only repair

Source field:

\$source_text

Prior translation:

\$raw_response

Parser errors:

\$errors

Preserve every valid claim, case, relation, expression, commitment, attribution,
 ↪ and endpoint. Do not improve the reasoning, add missing meaning, insert
 ↪ premises, or remove logically dubious relations. Correct only listed syntax,
 ↪ exact-span, or reference failures. If an invalid relation cannot satisfy
 ↪ containment without changing endpoints, omit that relation but keep all claims.

Every T0 has one claim ID.

Return the complete repaired translation only, without Markdown or explanation.

491 **I.7 Relation format repair: system instruction**

Perform format-only repair of relation records. Claims and substantive relation
 ↪ choices are immutable.

492 **I.8 Relation format repair: user template**

Relation format-only repair

\$relation_instructions

Prior relation output:

\$raw_response

Parser errors:

\$errors

Preserve every valid relation exactly. Add only repaired `eN` IDs from rejected
↪ lines. Do not add claims, cases, new meaning, hidden premises, or missing
↪ relations.

Return the complete repaired relation output only, without Markdown or explanation.

493 J Limitations

494 Development evaluation found 94.7% claim precision and 97.0% commitment/attribution accuracy
495 on matched expressions, with 76.5% full claim recall. Targeted source review corrected additional
496 errors, and the affected results were recomputed. These checks provide evidence that extracted claims
497 are generally faithful on the evaluated sample, while the recall results indicate incomplete coverage.
498 A larger held-out evaluation would be needed to estimate translation error rates across the full corpus.
499 We analyze what agents say, not their internal reasoning, and an omitted reasoning step may still have
500 occurred. Uptake cannot establish peer influence. AUC does not establish causation or predictive
501 value beyond initial accuracy. The 120 debates cover only 40 distinct puzzles, with few weak-team
502 successes and strong-team failures. Generalization beyond KKS remains uncertain. Appendix G
503 describes the translation review and sensitivity analyses.

504 K Broader impacts

505 These tools could help researchers design debate protocols that encourage useful information sharing
506 and identify persistent errors. They could also help people review multi-agent explanations by show-
507 ing which claims were made, how they were supported, and where agents disagreed. As agent sys-
508 tems become more capable and interconnected, interpreting their interactions may help researchers
509 identify how errors spread or persist and assess safety concerns that arise from collective behavior.
510 Such analysis supports investigation; it does not establish that a system is safe. This study analyzes
511 synthetic logic puzzles and introduces no deployed system or trained model; we do not identify a
512 direct negative societal application within its evaluated scope.

513 **NeurIPS Paper Checklist**

514 **1. Claims**

515 Question: Do the main claims made in the abstract and introduction accurately reflect the
516 paper’s contributions and scope?

517 Answer: [Yes]

518 Justification: The abstract and introduction describe the claim-level observability frame-
519 work and descriptive associations in the corrected 120-debate corpus. The abstract reports
520 the current AUC values, identifies the inconclusive association for uptake selectivity, and
521 does not claim incremental predictive value or causal influence. Section 3 reports the cor-
522 responding estimates and Appendix J states their limitations.

523 **2. Limitations**

524 Question: Does the paper discuss the limitations of the work performed by the authors?

525 Answer: [Yes]

526 Justification: Appendix J reports the translation evaluation and targeted corrections, ac-
527 knowledges incomplete claim coverage and the scope of the evaluation, and discusses re-
528 striction to communicated content, the gap between expressed and internal reasoning, up-
529 take’s inability to establish influence, the descriptive status of AUC, the 40 distinct puzzle
530 clusters (20 per size) and sparse outcome cells, and uncertain generalization beyond KKS.
531 Appendix G adds the attribution-audit and correction-overlay caveats.

532 **3. Theory assumptions and proofs**

533 Question: For each theoretical result, does the paper provide the full set of assumptions and
534 a complete (and correct) proof?

535 Answer: [N/A]

536 Justification: The paper states definitions and estimators, including an information measure
537 over an enumerated assignment space, but proves no theorem and claims no theoretical
538 result.

539 **4. Experimental result reproducibility**

540 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
541 perimental results of the paper to the extent that it affects the main claims and/or conclusions
542 of the paper (regardless of whether the code and data are provided or not)?

543 Answer: [Yes]

544 Justification: The accompanying archive contains all 120 raw debate outputs, benchmark
545 labels, configurations and prompts, six translation sources, the correction ledger and over-
546 lays, source-level scored data, scorer code, and bootstrap code. Its documented commands
547 were replayed successfully through correction integration, source-level rescoring, metric
548 regeneration, and puzzle-cluster intervals; Appendix E describes this scope.

549 **5. Open access to data and code**

550 Question: Does the paper provide open access to the data and code, with sufficient instruc-
551 tions to faithfully reproduce the main experimental results, as described in supplemental
552 material?

553 Answer: [Yes]

554 Justification: The anonymized supplementary archive is provided with the submission. Its
555 README gives exact dependency versions and reproduction commands, and its manifest
556 hashes the released files. The extracted package reproduced the corrected 120-debate tables
557 without new model calls. This answer assumes that the verified archive is attached.

558 **6. Experimental setting/details**

559 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
560 rameters, how they were chosen, type of optimizer) necessary to understand the results?

561 Answer: [Yes]

Justification: No model is trained; this is inference-only research. Appendix A reports the sampling temperature, reasoning-effort settings, agent count and ordering rule, debate depth, visibility conditions, aggregation and tie-breaking, output contract and format-repair policy, and the 20 shared puzzle identifiers for each size. Strength tiers were assigned from prior solo qualification on the same benchmark pool, which Section 2 states is not held-out model selection.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Tables 2 and 3 and the central strong-versus-weak uptake comparison report 95% percentile intervals from a puzzle-cluster bootstrap. Appendix B states the resampling unit, why it is the puzzle rather than the claim–recipient pair, the number of draws (10,000), the seed (20260906), the treatment of resamples with an empty denominator or a single outcome class, the number of defined draws, and the eligible sample for each metric, including why uptake metrics have 71, 77 or 96 eligible debates rather than 120. The revised main-text AUC intervals use one puzzle-cluster scheme throughout; earlier size-6 robustness outputs are explicitly labeled as historical. The intervals are described as exploratory and conditional on fixed rosters and translated claims.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix E and the supplementary COMPUTE.md document hosted inference providers, local and orchestration resources, model-call settings, recorded usage, and available execution timings; the archive provides the environment and commands needed to reproduce the reported analysis. Provider-side hardware is not disclosed, and the appendix explicitly identifies gaps in aggregate translation and semantic-review accounting.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The study uses procedurally generated synthetic puzzles and hosted model APIs, involves no human subjects or personal data, and discloses failures, exclusions, corrections, uncertainty, and limitations, including a disclosed translation-correction overlay and an inconclusive attribution audit.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix K discusses benefits for interpreting agent interactions, diagnosing error propagation, and studying future safety concerns. It explains the limited direct societal impact of the present synthetic-puzzle study and states that behavioral analysis does not establish system safety.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The planned release contains synthetic logic puzzles, prompts, model outputs on those puzzles, translation records, and analysis code. It introduces no trained model, no personal data, and no scraped dataset, and therefore poses no high risk of misuse requiring gated release.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Appendix F lists every model, serving provider, and material software dependency with its role and applicable license or provider terms, and distinguishes independently generated puzzles, transcripts, and analysis code from reused assets. The debate protocol and task family inherited from Wu et al. [2025] are credited in Section 2 and Appendix H. Hosted models are accessed as services under provider terms and are not redistributed.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The archive documents the fresh size-8 extension and corrected translations alongside the original size-6 assets. README, COMPUTE, schemas, raw-run records, correction ledgers, and before/after provenance describe their contents, intended use, generation, and limitations; Appendix E summarizes the release.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The study includes no crowdsourcing and no human subjects. The reference annotations used for translator development were produced by the authors with model assistance, not by external or paid annotators.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The study involves no human participants and no personal data, so no IRB or equivalent review applies.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are both the objects of study and a core methodological component. Appendix A declares the debater and supervisor models; Appendix C declares the two-pass translator, its reasoning-effort setting, its prompts, and its format-repair and cleanup models; and Appendix G declares the model-assisted review of sampled uptake pairs and the resulting polarity corrections.